

cs482-assn2

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A-score2 method description

-use ensemble of 1D CNN(Convolutional Neural Network) classifier and weighted kmer score

-CNN part:

-input: encode peptide sequence to integer, padding to data len 40

-structure:

embedding(64dim) -> 3 parallel Conv1d(kernel=3,5,7, filters=64) -> max pooling -> dropout(0.3) -> fully connected -> 2 type output

-train data:

SwissProt protein data natural peptides + shuffle generate random peptides

-weighted kmer part:

-use k=2,3,4 $\log_2(p/q)$ score tables with calibrated weights

-ensemble method:

z-score normalize CNN output and weighted kmer output, then mix 50/50 as final score2

B-test data description

-sampling 1000 natural peptides from swissprot original data

-shuffle character but total character not change to generate 1000 random peptides

-shuffle sequence of 1000+1000 data as input data

C-accuracy data on test file

-score1 accuracy: 0.6670

-score2 accuracy: 0.7960

-running time: 1.44 sec

-estimated memory: ~60MB